Shanshan Zhu

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EDUCATION

M.Sc. in Psychology, Psychology College, Nanjing Normal University Since 9/2023

Core Courses: Experimental Psychology, R Programming in Psychology, Advanced Research Methods

B.A. in Health Services and Management, Public health College, Hangzhou Normal University. 2019 – 2023

Core Courses: Medical Statistics, Epidemiology, Clinical Medicine, Advanced Mathematics C1

ACADEMIC EXPERIENCE

Master thesis: Self-Reference Modulates Stimulus Representations in Visual Cortex Since 12/2024

Role: Project leader; **Advisors:** Prof. Hu Chuan-Peng and Asso. Prof. Yan Chu-Yao

My work focuses on designing a behavioral and neuroimaging study to examine how self-related information influences early visual perception. I collect behavioral and magnetic resonance imaging (MRI) data from participants, perform data processing and behavioral analysis using R, and analyze functional MRI data using Python.

A Cognitive Ontological Meta Database for Self-Referential Processing Since 11/2023

Role: Project leader; **Advisor:** Prof. Hu Chuan-Peng

My contributions are expanded a structured database summarizing how different studies investigate self-related thinking in the brain. Created a codebook, reviewed scientific papers, extracted key information, and organized all materials for public sharing. Maintained documentation to ensure the dataset can be reused by other researchers.

Neuroimaging Meta-Analysis of Self-Referential Processing Across Psychiatric Disorders Since 3/2024

Role: Project leader; **Advisor:** Prof. Hu Chuan-Peng

Conducted a coordinate-based fMRI meta-analysis to identify altered neural responses to self-referential judgments in psychiatric disorders, including literature screening, data extraction, and Python-based analysis.

The Impact of Legal Professional Education on Moral and Legal Reasoning Since 5/2024

Role: major contributor; **Advisor:** Prof. Hu Chuan-Peng

Analyzed behavioral and brain-imaging data to understand how legal training influences decision-making. Conducted data cleaning, statistical modeling, and interpretation of results. Contributed to writing the research manuscript.

Chinese Child Brain Development Project Since 11/2024

Role: MRI Operator; **Advisor:** Prof. Hu Chuan-Peng

Operated MRI scans for children aged 6–8, including pre-scan familiarization to reduce anxiety, monitoring head motion, ensuring proper alignment, and acquiring resting-state fMRI, structural MRI, and diffusion-weighted imaging (DWI) data.

PUBLICATIONS

Chen, Y., Zhou, Y., Li, M., Hong, Y., Chen, H., **Zhu, S.**, Zhou, Y., Yang, S., Wu, X., & Wang, D. (2023). Social capital and loneliness among older adults in community dwellings and nursing homes in Zhejiang Province of China. *Frontiers in Public Health*, *11*. https://doi.org/10.3389/fpubh.2023.1150310

Cheng, H., Yang, Q., Wang, R., Luo, R., **Zhu, S.**, Li, M., Li, W., Chen, C., Zou, Y., Huang, Z., Xie, T., Wang, S., Zhang, H., & Tian, Q. (2022). Emerging Advances of Detection Strategies for Tumor-Derived Exosomes. *International Journal of Molecular Sciences*, *23*(2), 868. https://doi.org/10.3390/ijms23020868

Meng, F., Ye, M., Si, J., Chen, W., Hong, Y., Liu, S., Chen, Y., Shen, X., **Zhu, S.**, Zhao, C., Guo, M., Feng, X., & Wang, D.

(2022). Status of traditional Chinese medicine healthcare services in nursing homes across China. *Geriatric Nursing*, *45*, 93–99. https://doi.org/10.1016/j.gerinurse.2022.02.025

Zhu, S., and Hu, C. (2025). Shared brain basis for altered self-referential processing across psychiatric disorders? A systematic review and meta-analysis of neuroimaging studies. (in preparation)

SELECTED CONFERENCE PRESENTATIONS

Oral talk: 25th National Congress of Psychology Chengdu, China 10/2023

Zhu, S., Wu, J., Sun, S., Xiao, J., & Hu, C. (2023). Abnormal brain activity in self-referential processing among psychiatric patients: Evidence from an ALE meta-analysis. In Proceedings of the 25th National Congress of Psychology (pp. 486–488). (Published abstract; Basic Psychology category. This is the highest-profile and largest academic platform for psychological research in China)

Oral talk: Jiangsu Psychological Society Nanjing, China 11/2025

Zhu, S., Wu, S., Zhu, X., Fan, Z., & Hu, C. (2025). A neuroimaging meta-database of self-referential encoding paradigms and its applications in psychological research. (Basic Psychology category)

TEACHING EXPERIENCE

Bayesian Statistics in Psychology **Teaching Assistant** Autumn 2024

Instructor: Prof. Hu Chuan-Peng, Nanjing Normal University.

Prepared Python-based teaching materials and interactive exercises, and provided technical support to graduate students in data analysis and programming.

PROFESSIONAL TRAININGS

Research training, FMRIB Software Library (FSL) neuroimaging analysis, Institute of Psychology, Chinese Academy of Sciences, Beijing, China, 7/2024

Gained hands-on experience in using FSL software for neuroimaging data analysis, including preprocessing, statistical modeling, and visualization techniques.

Programming Training, Natural Language Processing, NetEase Intelligence, Hangzhou, China, 2023.05 – 2023.09

Engaged in NLP techniques for text analysis, including annotating and semantic processing using Python.

Research internship, Sir Run Run Shaw hospital, Hangzhou, China, 2022.07 – 2022.11

Assisted in the submission of grant proposals, including document preparation, proofreading, and coordination of required institutional approvals.

AWARDS & ACHIEVEMENTS

12/2023	First-Class Scholarship for Master’s Students, awarded to the top students in the cohort), Nanjing Normal University (¥ 12000)
6/2023	Outstanding Graduate Award, Hangzhou Normal University
3/2023	Hailiang Scholarship, one of 20 recipients, Hangzhou Normal University (¥ 5000)
6/2022	Third Prize, 2022 China Undergraduate Computer Design Competition (National-level competition)
5/2022	Silver Medal, 13th Zhejiang “Challenge Cup” College Student Entrepreneurship Plan Competition

SKILLS

Languages: Chinese (native); English (fluent); Russian (basic)

Programming: Python; MATLAB; R

Neuroimaging: Analyzing MRI data based on task; specialized neuroimaging software (e.g., FMRIB Software Library); designing and implementing psychological experiments in MRI

Tools: GitHub (software environment management); Docker (software environment management), Jupyter notebook, LaTeX (scientific writing), Microsoft Office